HireMind: Efficient CV–Job Fit Classification through Distillation and Quantization

Item 4 — Final Experimental Results

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Abstract

A compact model for matching a CV to a job must recognize suitable candidates as well as fit within a local inference budget. We compare supervised BERT-Tiny (M0), the same student distilled from a BERT-base teacher (M1), and a quantized student with compressed embeddings and Linear layers (M2). We select checkpoints on a job-disjoint validation partition, freeze the choices, and evaluate nine checkpoints across three paired seeds on 1,759 held-out pairs. Mean test macro-F1 is 33.70 for M0, 31.88 for M1 and 32.39 for M2 on a percentage scale. Distillation misses the prespecified one-point improvement. Quantization reduces serialized state size by 73.38% and increases mean macro-F1 slightly relative to M1, while model-only CPU latency rises from 5.56 to 6.31 milliseconds. In one seed M2 identifies only seven of 458 suitable pairs. These findings support storing a smaller classifier, but the variation in class recall limits its use for hiring decisions. The proposed Senior workflow retains recruiter oversight and requires separate deployment validation.

1 Introduction

HireMind is the Senior project's recruitment platform. Its planned workflow includes CV processing, job matching, weighted scoring, practical verification and recruiter-reviewed feedback. A small local classifier could provide a text-compatibility signal within that workflow. This DL study evaluates that possibility without claiming that the current Senior prototype already runs the model.

Our contribution is a paired comparison of supervision, teacher guidance and component-wise quantization on frozen recruitment data, including direct CPU measurements and class-level errors. The research question is: *Can teacher-guided training improve a compact CV–job fit classifier, and can post-training quantization reduce storage while retaining useful predictive performance?* Local inference could reduce dependence on repeated external calls; cost, privacy and deployment benefits still require an implemented comparison. Label prediction is not evidence that a candidate possesses a skill.

2 Related Work

The earlier literature review is reused. BERT supplies the encoder architecture (Devlin et al., 2019); compact pretrained students provide a practical starting point for distillation (Turc et al., 2019). Soft-target distillation transfers class relationships from a teacher (Hinton et al., 2015). DistilBERT and TinyBERT illustrate compression through distillation, but use training objectives and settings that differ from our task-specific logit loss (Sanh et al., 2019; Jiao et al., 2020).

Q8BERT examines 8-bit BERT, while TernaryBERT combines distillation and lower precision (Zafrir et al., 2019; Zhang et al., 2020). Transformer quantization can be sensitive to activation distributions (Bondarenko et al., 2021). These studies motivate measuring both predictive quality and actual latency, rather than inferring speed from storage. The recruitment framework of Gan et al. (2024) provides application context, not a directly comparable benchmark. We retain the nine references from Item 2; their reported benchmark scores are not treated as HireMind results.

3 Dataset and Scope

The approved course dataset is `cnamuangtoun/resume-job-description-fit`. Its original 8,000 pairs contain 643 CVs and 351 jobs. The prior audit removed 12 rows from six conflicting pairs, one same-label duplicate, and 73 rows associated with near-duplicate jobs. The resulting frozen split is reused exactly. Job groups are disjoint across partitions, but 474 CVs overlap between training and test. Consequently, the intended gen-

Partition	Pairs
Training	4,993
Validation	1,162
Held-out test	1,759

Table 1: Frozen Item 2 partitions. Test data were reserved until model selection was frozen.

eralization claim is unseen jobs with potentially familiar CVs, not unseen candidates. Dataset source provenance, licence and label-generation method remain undocumented; instructor approval does not resolve these limitations.

The development runner downloads the pinned training CSV, verifies its SHA-256, and applies the frozen row IDs. Text is tokenized in full before longest-first truncation to a combined length of 256, with no earlier character cutoff. This may discard relevant CV evidence. Longer inputs or section selection belong to a separate controlled follow-up, not a silent change to this comparison.

4 Models and Hypotheses

M0. The student is `google/bert_uncased_L-2_H-128_A-2`, with two layers, hidden width 128, two attention heads and a three-class head. Its 4,386,307 logical parameters include the large embedding table. The checkpoint revision is pinned in code. M0 uses cross-entropy.

M1. The same student is trained with

$$L = 0.5L_{\mathrm{CE}} + 0.5T^2\mathrm{KL}(p_T \| p_S), \qquad (1)$$

where teacher and student probabilities use temperature T. A BERT-base teacher is fine-tuned only on the training partition, selected by validation macro-F1, and frozen. Its training logits are cached in evaluation mode. No validation or test labels enter the distillation targets. A weak teacher is reported as a limitation; the method is not replaced after seeing an unfavourable result.

M2. M1 is converted on CPU using dynamic INT8 Linear layers and 8-bit weight-only embeddings with floating per-row quantization parameters. LayerNorm remains FP32. There is no additional training. The same scalar weights remain logically present; quantization changes their representation, not the architectural parameter count or dense MAC estimate. M1 already has M0's small architecture, so distillation alone does not reduce M0's inference cost and incurs extra teacher training.

Frozen final hypotheses. H1 requires mean test macro-F1 of M1 to exceed M0 by at least one percentage point. H2 requires M2's saved state size to be at most 40% of M1's, a macro-F1 loss of at most one point, and retention of at least half the M1 gain over M0. These are final-test hypotheses. Item 3 validation observations are provisional and cannot confirm them. If H1 fails, gain retention is exploratory; storage and quality loss remain interpretable.

5 Experimental Setup

5.1 Controlled training

M0 and M1 share the pretrained initialization, paired head seed, data order, input length, optimizer and update budget. AdamW uses weight decay 0.01, gradient clipping 1.0, linear decay with 10% warmup, effective batch size 16 and three epochs (939 updates). Validation runs every 100 updates and at the end; maximum validation macro-F1 chooses a checkpoint. Full-precision eager attention is used.

M0 searches four learning rates: 2×10^{-5}, 5×10^{-5}, 10^{-4} and 3×10^{-4}. M1 searches $\{5 \times 10^{-5}, 10^{-4}\} \times T \in \{2, 4\}$, also four trials. Search uses seed 42; selected settings repeat with seeds 43 and 44. Thus inference comparisons use three paired student seeds. The single teacher searches two learning rates, 2×10^{-5} and 5×10^{-5}. Teacher micro-batches of four accumulate into the same effective batch of 16 to fit GPU memory. This difference is explicit; student micro-batches are 16. Teacher seed variation is not studied.

Validated completed M0 trials are reused from the prior run folder. Historical Item 2 software and timing are retained separately; deployment costs for all models are remeasured in the same CPU benchmark session. Training used a Tesla T4, Python 3.13.16, PyTorch 2.11.0+cu130, CUDA 13.0, Transformers 4.51.3, NumPy 2.1.3 and pandas 2.2.3. CPU benchmarking used an Intel Xeon CPU reported at 2.00 GHz. Seeds are 42, 43 and 44. Deterministic settings reduce avoidable variation but do not imply equivalence across software or hardware releases.

A storage incident removed several checkpoint files. Valid completed trials were reused; missing student trials were repeated under the same protocol. The selected Teacher was resumed from verified update 600 to update 939. Its recovered model reproduces the original cached training log-

its exactly for all 4,993 inputs (maximum absolute difference 0.0). The serialized SHA-256 differs; original records and targets are preserved, with a separate recovery record and comparison. This establishes agreement on those inputs, not byte-identical weights or universal prediction equivalence. The saved CPU incident audit verified the recovered Teacher and its original cached targets.

5.2 Efficiency measurement

All deployment timings use the same CPU, batch size one, one PyTorch thread and sequence length 256. Each model/seed is measured in a fresh process using the same 20 fixed validation inputs, 30 warmups and 200 timed calls. We report median and IQR, retaining Q1, Q3, p95 and every sample. Model-only timing excludes tokenization; tokenization-inclusive timing is recorded separately. Runs are sequential to avoid concurrent benchmarking.

Size means serialized model state only, excluding tokenizer, configuration and optimizer. Tables use decimal MB; CSVs also report MiB. Logical parameter counts include packed quantized weights. The analytical dense MAC estimate is 134,234,496 per pair; it excludes non-matrix operations and is not a measured instruction count. Multiplying by two gives a conventional approximate matrix-operation FLOP count, not total execution cost.

Peak process RSS includes imports, loading, tokenization, inference and quantization conversion/reload. It is not isolated inference memory; conversion can inflate M2's peak. GPU training peak allocation and training wall time are recorded separately. We do not claim an inference-memory reduction from this broader process measure.

6 Main Comparison

Table 2 summarizes selected checkpoints across the three paired student seeds; all quality values are validation percentages. Means and sample standard deviations describe variation over seeds, not confidence intervals. Equal search-trial counts reduce one source of selection imbalance, but different learning-rate spaces and the additional Teacher search remain explicit design choices.

The selected learning rates are 10^{-4} for M0 and 5×10^{-5} for M1; M1 uses $T = 2$. M1–M0 macro-F1 differences for seeds 42, 43 and 44 are respectively -0.20, -0.38 and -7.61 percentage points, a mean difference of -2.73. M1 therefore offers no

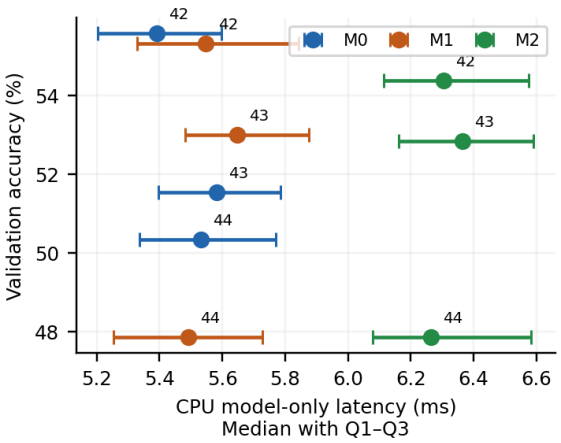


Figure 1: Accuracy versus CPU model-only latency. Labels identify student seeds. Horizontal bars are Q1–Q3 of 200 timed calls, not confidence intervals. M2 is slower than M1 in every paired seed on this machine.

macro-F1 improvement in any of these three paired runs. Its seed-43 accuracy improves while macro-F1 decreases, illustrating why accuracy alone does not establish a better balanced classifier.

The single selected Teacher attains 50.46 validation macro-F1 and 56.11% accuracy. It exceeds M0 seed 42 by 3.93 macro-F1 points but is not a highly accurate reference model. Teacher strength is a possible limitation; these observations do not isolate the cause of the student result. We did not run three independent Teachers.

7 Ablation Study

7.1 Quantized components

The Linear-only, embeddings-only and combined variants use the same selected M1 weights for each seed. Mean macro-F1 is 37.20, 37.10 and 36.96 respectively, compared with 37.28 for FP32 M1. Serialized states occupy 16.342, 5.894 and 4.675 MB respectively. The embedding table accounts for most of the observed storage reduction. Combined quantization gives the smallest state but the largest mean quality loss, 0.32 percentage points. Its prediction agreement with M1 is 96.56%, 97.76% and 98.71% for seeds 42–44. Agreement with another model is not accuracy against ground truth.

7.2 Temperature

An additional $T = 1$ distillation ablation holds the selected learning rate, loss mixing weight, update budget and all other student settings fixed. Table 4 reports each trial's best validation checkpoint, not necessarily its final update. Softening helps seed 42

Model	Accuracy (%)	Macro-F1 (%)	Params (M)	State (MB)	MACs (G)	CPU median (ms)
M0	52.50 ± 2.75	40.01 ± 5.72	4.386	17.559	0.134	5.502 ± 0.099
M1	52.07 ± 3.83	37.28 ± 9.05	4.386	17.559	0.134	5.563 ± 0.079
M2	51.69 ± 3.42	36.96 ± 8.86	4.386	4.675	0.134	6.311 ± 0.050

Table 2: Validation comparison. Quality and latency columns show mean $\pm$ sample SD over three seeds. Latency aggregates each seed's 200-call median; per-run IQRs appear in Table 3. State size is the mean serialized state in decimal MB, excluding tokenizer, configuration and optimizer. Logical parameters and analytical dense MACs are unchanged by quantization.

Model	Seed	Median ms	IQR ms	Peak RSS MiB
M0	42	5.391	0.392	1507.9
M0	43	5.582	0.389	1591.1
M0	44	5.532	0.435	1588.2
M1	42	5.548	0.514	1589.4
M1	43	5.648	0.394	1588.4
M1	44	5.492	0.474	1589.3
M2	42	6.305	0.461	1593.5
M2	43	6.364	0.430	1594.1
M2	44	6.265	0.505	1593.5

Table 3: Individual CPU runs. Peak RSS is whole-process peak, including imports, loading, conversion where applicable, and inference. It does not isolate inference-only memory.

Seed	$T = 1$ macro-F1	$T = 2$ macro-F1
42	44.21	46.34
43	38.52	37.26
44	30.13	28.25
Mean	37.62	37.28

Table 4: Temperature ablation. $T = 2$ was selected during the original validation search.

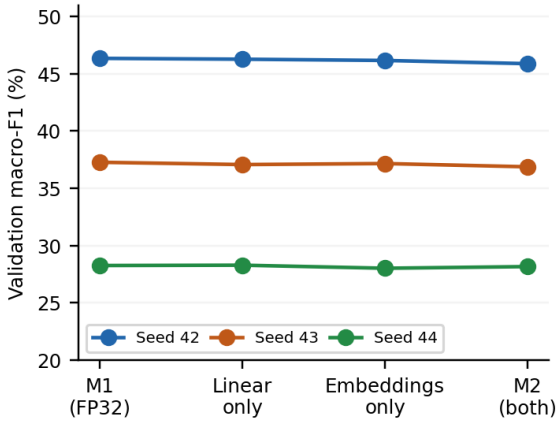


Figure 2: One-factor quantization ablation with paired seeds. The between-seed spread is much larger than the within-seed change induced by each quantization choice. Lines connect the same underlying student checkpoint.

but not seeds 43 or 44. The small mean difference does not establish a reliable temperature advantage. The selected $T = 2$ model is not silently replaced after inspecting this exploratory ablation. Quantizing M0 remains an optional future comparison, not a completed result or a requirement added to this milestone.

8 Efficiency Analysis

Relative to M1, M2's mean serialized state is 73.38% smaller, approximately 3.76 times smaller. This is a model-state storage result; it is not a 73% reduction in the full application, parameters, MACs, peak RAM or training cost. The mean of per-seed median CPU latencies rises from 5.563 to 6.311 ms, an increase of 13.46%. All three paired runs show this direction. Tokenization-inclusive medians are also retained in the CSV evidence; their measurement scope differs from the model-only plot.

Quantization overhead and limited opportunities to accelerate a tiny network are plausible explanations for slower inference. No operator-level profile was run, so these are hypotheses rather than established causes. The process peaks of roughly 1.5–1.6 GiB are dominated by a broader execution scope than the model state; this measurement does not demonstrate lower inference RAM. PyTorch's deprecation warnings were recorded, and the implementation is tied to the tested software version rather than silently migrated during the experiment.

Distillation does not shrink M0's architecture because both M0 and M1 already use the same BERT-Tiny student. It also requires Teacher training and target generation. The recovered selected Teacher record reports 1,718.94 seconds and a peak allocated GPU memory of 2,788.53 MiB. Its elapsed-time scope includes recovery I/O and excludes unsaved interrupted work; it is neither the total project GPU cost nor a fair isolated training-speed comparison. We report no energy or monetary savings.

9 Held-out Results

After freezing the selected checkpoints and verifying their identities, we evaluated the same 1,759 held-out pairs for each model and seed. The pinned test file, text identifiers and row order were checked before scoring. Test results were used only for reporting. All nine confusion matrices and scores were independently recomputed from saved predictions and fixed labels. H1 fails: mean M1 minus

	Accuracy	Macro-F1	Good Fit recall
M0	45.33 ± 1.12	33.70 ± 3.92	16.16
M1	47.19 ± 0.82	31.88 ± 5.07	12.52
M2	47.60 ± 0.61	32.39 ± 5.23	12.81

Table 5: Held-out percentages over three paired seeds. Dispersion is sample SD. Recall is averaged across seeds.

M0 macro-F1 is -1.82 points against a required gain of at least one. For H2, the M2/M1 state-size ratio is 0.266 for every seed, below the 0.40 ceiling. Mean M2 minus M1 macro-F1 is $+0.51$ points, meeting the one-point maximum-loss criterion. The gain-retention criterion has no confirmatory interpretation because the prerequisite positive M1 gain was absent. The two met criteria do not establish H2 in full. These paired comparisons are descriptive. At seed 44, M2 identifies seven of 458

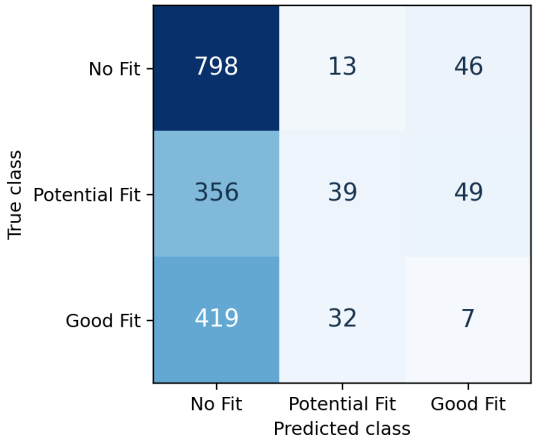


Figure 3: Held-out M2, seed 44. Rows are true classes and columns predicted classes. Counts describe one checkpoint, selected here to illustrate a failure.

Good Fit pairs and assigns 419 to No Fit. M1 already has only 1.97% Good Fit recall at this seed. The failure therefore predates quantization. The held-out error counts describe model behaviour; they cannot identify the cause of a particular mismatch without examining the text.

10 Discussion

The observed benefit is reduced model-state storage at a modest validation macro-F1 loss. Distillation does not provide the intended quality improvement, and quantization does not provide CPU speedup on the measured hardware. Consequently, M2 cannot be described as preserving a positive M1 advantage over M0: there is no positive mean advantage in this validation comparison. The held-out evaluation below tests the frozen hypotheses.

TinyBERT and DistilBERT demonstrate that distillation can transfer useful representations in other settings (Jiao et al., 2020; Sanh et al., 2019). Our small student uses task logits from one fine-tuned teacher and shows no mean test gain. Different datasets, objectives and training budgets preclude a direct score comparison. Our observed size reduction is consistent with the compression goal of Q8BERT (Zafrir et al., 2019); the latency measurements show why a deployment claim also requires testing the target CPU.

Seed sensitivity is substantial on validation. Good Fit recall for M1 is 30.80%, 32.87% and 0.69% across seeds 42–44; M2 yields 31.14%, 30.45% and 0.35%. The seed-44 model nearly misses that class altogether. Confusion matrices and class reports expose this weakness even though overall accuracy is close to 48%. The figures support caution about reliability and preclude an automated rejection claim. They do not alone establish why examples are misclassified.

Possible explanations include limited Teacher quality, sensitivity to initialization and data order, class imbalance, truncation and uncertain labels. These have not been experimentally isolated. After training, both authors independently assessed the same 90 training pairs and reviewed validation errors. These post-hoc reviews did not change the labels used for training. We report no numerical agreement statistic.

10.1 Limitations

Repeated validation selection can produce optimistic development scores. Other limits include three student seeds, one Teacher, recurring CVs, uncertain data provenance and one CPU benchmark environment. This study does not demonstrate generalization to unseen candidates, calibration as a hiring probability, fairness across demographic groups or readiness for production deployment. Negative findings are retained rather than

changing hypotheses after observing results.

11 Connection to Senior

The recruitment problem, terminology and workflow are reused from Senior. The compact classifier, validation protocol and efficiency measurements are new DL work. A successfully evaluated checkpoint could later be integrated into the backend as an advisory matching signal. It does not currently replace the planned API, implement practical verification, learn automatically from database storage, or justify autonomous rejection. Integration and a comparison against the actual Senior matching approach are future work.

12 Conclusion

The compact quantized state satisfies the storage target on every seed. The supervised baseline achieves the highest mean held-out macro-F1, while the quantized student is slower on the measured CPU. A subsequent study could test longer inputs on a new evaluation cohort and compare the advisory model with the Senior workflow under recruiter review. The 512-token setting remains untested.

Author Contributions

Hams Aljohani audited and cleaned the dataset, built the job-grouped train/validation/test split, implemented and trained the M0 baseline (learning-rate search, overfit-one-batch test, three seeds), measured its CPU cost, and set up the project repository. Rawan Khojah fine-tuned and recovered the BERT-base teacher, trained the distilled M1 students, built and benchmarked the quantized M2 variants, and ran the frozen held-out evaluation. Both authors independently reviewed the same 90 training pairs, examined validation errors, and prepared the paper, slides and poster together.

AI Assistance and Author Review

ChatGPT and Claude were used for code debugging and language editing. All experiments, results and conclusions were produced and verified by the authors.

Reproducibility

The repository contains the training and evaluation notebooks, the frozen data splits, the experiment settings, the validation and test results, the CPU benchmark records and scripts that reproduce the tables and figures. Code and results: `https://github.com/RAWANKHOJAH/HireMind-DL` (folders `assignment_3` and `assignment_4`).

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